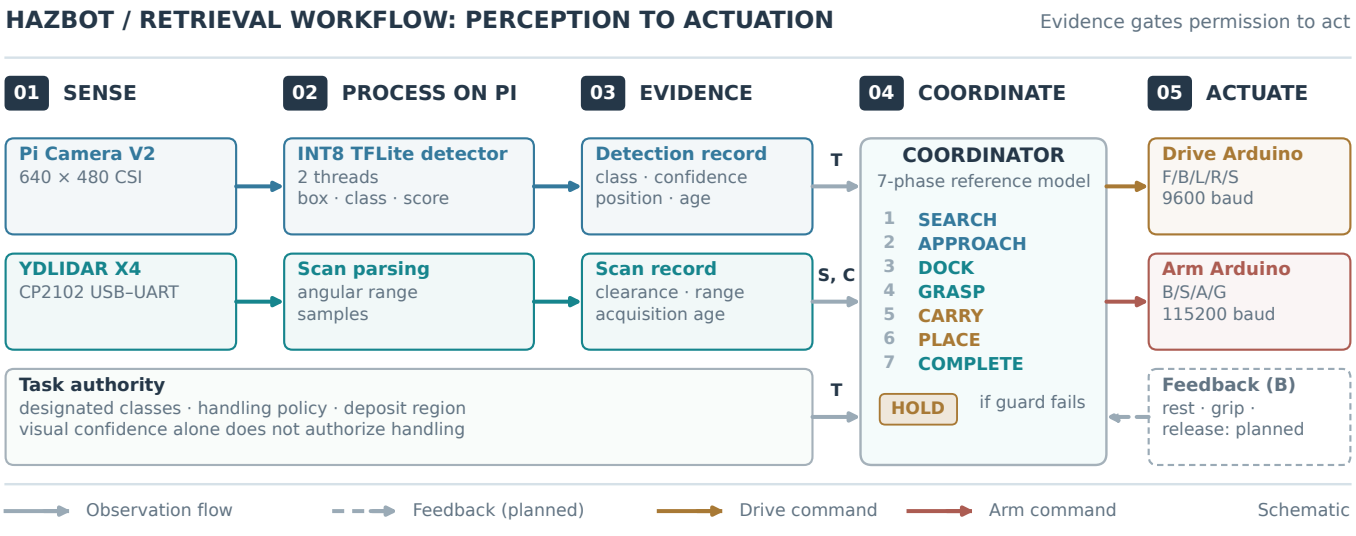
HAZBOT / RETRIEVAL WORKFLOW: PERCEPTION TO ACTUATION
Evidence gates permission to act
01 SENSE
02 PROCESS ON PI
03 EVIDENCE
04 COORDINATE
05 ACTUATE
Pi Camera V2
640 × 480 CSI
YDLIDAR X4
CP2102 USB–UART
INT8 TFLite detector
2 threads
box · class · score
Scan parsing
angular range
samples
Detection record
class · confidence
position · age
Scan record
clearance · range
acquisition age
Task authority
designated classes · handling policy · deposit region
visual confidence alone does not authorize handling
T
S, C
T
COORDINATOR
7-phase reference model
1 SEARCH
2 APPROACH
3 DOCK
4 GRASP
5 CARRY
6 PLACE
7 COMPLETE
HOLD
if guard fails
Drive Arduino
F/B/L/R/S
9600 baud
Arm Arduino
B/S/A/G
115200 baud
Feedback (B)
rest · grip ·
release: planned
Observation flow
Feedback (planned)
Drive command
Arm command
Schematic